



BICOL UNIVERSITY
COLLEGE OF SCIENCE
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Individual Journal

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Group Name: Project Bloom

Role in the Group: Client/Frontend Developer, Database Administrator, Tester

Topic / Activity: Laboratory 4: REST API

Date Submitted: April 16, 2026

Overview of My Role

I served as the Client/Frontend Developer, Database Administrator, and Tester in our project. As a frontend developer, I was responsible for designing and developing the web-based client interface using HTML. As a database administrator, I handled the creation and management of the database used by the API. Additionally, I also assisted in performing tests to ensure the system worked correctly and handled errors properly.

Specific Tasks Performed

- Designed and developed the HTML interface for the client-side application so the system can be used in a web browser.
- Designed and managed the database (tables, fields, relationships)
- Ensured proper data storage and retrieval for API operations
- Conducted system testing, including checking API responses and validating inputs/outputs
- Identified and reported bugs or errors during testing

Code Contributions

I helped in developing the frontend interface using HTML, allowing users to interact with the system through a web browser. I created forms and input fields that collect user data and send requests to the REST API. I also handled the display of API responses, ensuring that outputs are clear and viewed in a readable format.

For the database, I created tables and defined the necessary fields required for the system. I ensured that the database works properly with the backend, allowing the API to retrieve, store, and update data correctly. I also helped in validating whether the data being stored and retrieved in the database was accurate.

In testing, I conducted tests to check if the system returned correct responses and handled errors properly such as missing or invalid inputs.

Challenges Encountered

One challenge I encountered during the project was managing the connection between the frontend and backend, especially when errors occurred. Handling issues like incorrect data retrieval or failed API requests was also challenging. There were also some issues with properly handling JSON responses and displaying them in a user-friendly way.

Another challenge was ensuring that the database schema supported all the required functionalities of the system while keeping it organized.

I also faced difficulties during testing, particularly in identifying errors coming from either the frontend, backend, or database.

Problem-Solving and Contributions

To solve these challenges, I carefully reviewed how API requests and responses work and adjusted the frontend to correctly process JSON data. I also improved the layout to make the output more readable.

For database issues, I revised the table structure and ensured that the fields matched the API's expected inputs and outputs.

During testing, I collaborated with my groupmates to trace errors and determine whether they originated from the frontend, backend, or database. I helped suggest fixes and retested the system to confirm improvements.